

Stop Saying Who, Start Saying How: A Mechanistic Approach to Public Health Messaging

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The term *men who have sex with men* (MSM) was adopted in the 1990s to reduce stigma by focusing on behavior over identity in HIV/STI prevention. Yet, its continued use has reinforced identity-based risk profiling, obscuring shared transmission mechanisms and limiting public health impact. Rising gonorrhea rates now span diverse populations — including women, heterosexual men, youth, and racial minorities — yet messaging often remains targeted to MSM, missing opportunities for broad prevention.

We propose a shift to a mechanistic, universal precaution framework that emphasizes *condomless penetrative sex* as the primary risk factor, regardless of gender or sexual identity. This approach treats risk as a function of behavior, not identity, aligning messaging with actual transmission dynamics. Using gonorrhea as a model pathogen — due to its high incidence, short incubation, and clear route of transmission — we test whether behavior-based messaging increases prevention uptake across all groups.

In a cluster quasi-experimental trial across 20 U.S. sexual health clinics (2025–2026), intervention sites will implement mechanistic messaging; control sites will maintain standard, identity-targeted campaigns. The primary outcome is adjusted gonorrhea positivity (cases per test) to account for testing-rate bias. Secondary outcomes include self-reported condom use via validated surveys and perceived stigma. Exploratory analyses will examine trends in antibiotic resistance markers, if available, to rule out treatment failure as a driver of changes.

We will use difference-in-differences analysis with time-series adjustments (e.g., for concurrent public health interventions) to compare pre- and post-intervention changes across arms. Subgroup analyses by gender, sexual orientation, and race/ethnicity will assess equity in impact without presuming risk.

Power analysis (G*Power): Assuming a baseline positivity rate of 5%, ICC = 0.02, 50 patients per cluster, and 20 clusters (10 per arm), we achieve 80% power ($\alpha = 0.05$) to detect a 30% reduction in positivity (to 3.5%) — a meaningful public health effect.

If effective, this behavior-centered model could replace identity-based STI messaging with a scalable, stigma-reducing framework. Because transmission depends on *acts*, not identities, this approach may be adapted across STIs, offering a pathogen-agnostic strategy for equitable, evidence-based public health communication